

An elementary endpoint distribution-function bootstrap

Dependency simplification: independent mathematical reviews completed

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Status. Independently mathematically reviewed, with no gap identified; not formally verified. This note replaces the classical localized-Laguerre input previously used for the exponent-3/4 distribution-function estimate. It uses only the reviewed Gaussian comparison and subset estimates. It does not assume the quadratic per-die theorem.

Use the notation of `../power_per_die_lower_bound.tex`: $u_j = 1 - p_j$, $s = \sum_j u_j$, $\ell = \lfloor m/4 \rfloor$, $N = m - \ell$, $\alpha = \ell/m$, $x = \alpha s$, and $\theta = 1/16$. For a uniform ℓ -subset B , write $A_B = \prod_{j \in B} p_j$, $\bar{A} = \mathbb{E}_B A_B$, and $v_B = N^{-1} \sum_{j \notin B} \ell u_j$. Let μ have row masses $(\ell + 1)w\bar{A}$, let ν have row masses mw , and let β have density $b_\ell(z) = (\ell + 1)\ell^{-1}(1 - z/\ell)^\ell$ on $[0, \ell]$. All constants are absolute.

Theorem 1. *For every monotone tensor with the exact uniform mixed moments,*

$$\nu([0, t]) \leq C(t + m^{-3/4}) \quad (t \geq 0).$$

The reviewed comparison supplies a starting estimate

$$\nu([0, t]) \leq C(t + h), \quad h = m^{-1/2}. \quad (1)$$

Here is a direct derivation, to specify the input precisely. Put

$$T_d(y) = \sum_{j < d} (1 - j/d) L_j(y) = d^{-1} L_{d-1}^{[2]}(y), \quad U_d(z) = d^{-2} T_d(\theta z)^2.$$

Its Laguerre coefficient product is bounded by an absolute constant. The reviewed bilinear comparison gives $\mu(U_d) \leq \beta(U_d) + Cd/m \leq C/d + Cd/m$ for $d \leq \sqrt{m}$. For $0 \leq x \leq c_0/d$, $U_d(x) \geq c > 0$ by the elementary near-zero Laguerre estimate. Choose c_0 so small that $x \leq c_0$ implies $s \leq 1/2$; then $\bar{A} \geq 1 - s \geq 1/2$ and ν is bounded by a constant times μ there. Taking $d \asymp \min(\sqrt{m}, 1/t)$ proves (1) near zero. For larger t , the finite row-mass estimate $\nu([0, t]) \leq C(t + 1)$ suffices.

Lemma 1. *Uniformly for $m^{1/4} \leq d \leq m^{3/4}$,*

$$|\mu(U_d) - \beta(U_d)| \leq C \left(\frac{1}{m} + \frac{hd}{m} + \frac{\log d}{dm} \right) + O(m^{-3}).$$

Proof. The Gaussian off-diagonal inequality states $y^{a/2} |L_r^{[a]}(y)| \leq e^{y/2} \sqrt{(r+a)!/r!}$. Since

$$D_z^j T_d(\theta z) = (-\theta)^j L_{d-j-1}^{[j+2]}(\theta z)/d,$$

the total matrix index is always $d + 1$. It follows, for every derivative order j simultaneously, that

$$|D_z^j T_d(\theta z)| \leq C e^{\theta z/2} z^{-1} (C \sqrt{d/z})^j.$$

Thus, for all $z > 0$ and all k ,

$$|U_d^{(k)}(z)|/k! \leq \frac{Ce^{\theta z}}{d^2 z^2} \frac{(C\sqrt{d/z})^k}{k!}. \quad (2)$$

The constants are uniform in k . The elementary whole-series bound

$$\sum_{k \geq 2} (k+1)a^k k^{k/2}/k! \leq Ca^2 e^{Ca^2} \quad (3)$$

follows by splitting even and odd orders and comparing to exponential series; for example $(2j)^j/(2j)! \leq 2^j/j!$.

We use only the older balance $\max u_j \leq C(s+1)/m$. The centered variance of $z_j = \ell u_j$, $j \notin B$, is at most $CN(s+1)v_B$. The centered elementary-coefficient bound therefore cancels the factors $v_B^{-k/2}$ in (2). Under the success-weighted subset law, the reviewed bias and second moment estimates are

$$0 \leq \mathbb{E}\Delta \leq Cx(x+1)/m, \quad \mathbb{E}\Delta^2 \leq C\{x(x+1)/m + x^2(x+1)^2/m^2\}, \quad \Delta = v_B - x, \quad (4)$$

when $s \leq m/256$. Also $\bar{A} \leq e^{-x}$.

For $x \leq 1/d$, use the global Gaussian derivative bounds with coefficient product $O(1)$. The complete polarization series and the singular Taylor averaging estimate each have pointwise error $O(d/m)$, because $s+1 = O(1)$. They vanish if $x = 0$. Their integrated contribution, by (1), is

$$O((d/m)(1/d+h)) = O(1/m + hd/m). \quad (5)$$

For $1/d < x \leq d$, first restrict to subsets with $v_B \geq x/2$. We always have $v_B < 2x$. The variance estimate, (2), and (3) bound the complete polarization error by

$$\frac{Ce^{\theta v_B}}{d^2 x^2} \frac{d(s+1)}{m} \exp[Cd(s+1)/m].$$

After averaging with A_B , $\bar{A} \leq e^{-x}$ leaves a fixed exponential margin, since $d/m \leq m^{-1/4} \rightarrow 0$. The resulting envelope is

$$\frac{C}{dm} e^{-cx} (x^{-1} + x^{-2}). \quad (6)$$

For averaging, retain the full linear bias and split only the Taylor remainder into good and bad subsets. On good subsets, use $|U_d'| \leq Cd^{-3/2}x^{-5/2}e^{2\theta x}$, $|U_d''| \leq Cd^{-1}x^{-3}e^{2\theta x}$, and (4). The same envelope results: the ratio of the linear term to the leading quadratic term is at most $C\sqrt{x/d} \leq C$, and the squared-bias term is absorbed in the exponential margin.

Here the bad-subset contribution is negligible. The uniform law satisfies

$$\Pr_B(v_B < x/2) \leq e^{-cm \min(x,1)}.$$

Indeed the entire population z_j has mean x and maximum $R = C(s+1)$; Maclaurin's inequality gives $\mathbb{E}e^{-Nv_B/R} \leq \exp[-Nx(1-e^{-1})/R]$, and Markov's inequality gives the claim. The full Gaussian series and the coarse Taylor remainder are polynomial in m times e^{Cx} on $x \leq d$. For $1/d < x \leq 1$, the bad exponent is at most $-cm/d$; for $1 \leq x \leq d$, it is at most $-cm + Cd$. Both are superpolynomially small uniformly for $m^{1/4} \leq d \leq m^{3/4}$. Weighted normalization causes no loss, since $\bar{A} \Pr_{\text{weighted}}(\text{bad}) = \mathbb{E}_{\text{uniform}} A_B \mathbf{1}_{\text{bad}} \leq \Pr_{\text{uniform}}(\text{bad})$.

For $x > d$, the global Gaussian series, $\bar{A} \leq e^{-x}$, and $d/m = o(1)$ give a polynomial factor times e^{-cx} . The nonsingular first derivative and uniform-subset Cauchy-Schwarz give the same estimate for averaging. These tails are $O(m^{-3})$ because $d \geq m^{1/4}$.

Finally (1) and Stieltjes integration by parts imply

$$\int_{1/d}^{\infty} e^{-cx} x^{-1} d\nu \leq C(\log d + hd), \quad \int_{1/d}^{\infty} e^{-cx} x^{-2} d\nu \leq C(d + hd^2).$$

Possible atoms at the cutoff are included by the same CDF bound. Integrate (6), add (5) and the negligible tails, and use the exact identity $\beta(P) = (\ell + 1)\mathbb{E}_w\mathbb{E}_B[A_B \text{Pol}_N(P)]$. This proves the lemma. $\square$

Proof of Theorem 1. For every $m^{1/4} \leq d \leq m^{3/4}$, Lemma 1 gives

$$\mu(U_d) \leq C/d + C\{1/m + m^{-1/2}d/m + \log d/(dm)\} + O(m^{-3}) \leq C/d.$$

The last inequality uses $d^2 \leq m^{3/2}$. Near-zero positivity and comparison of μ with ν , as in the derivation of (1), give $\nu([0, c_0/d]) \leq C/d$. For $t \lesssim m^{-1/4}$ choose $d \asymp \min(m^{3/4}, 1/t)$; for larger t use (1). This proves $\nu([0, t]) \leq C(t + m^{-3/4})$ uniformly. $\square$